



Forecasting movements of stock time series based on hidden state guided deep learning approach

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ABSTRACT

Stock movement forecasting is usually formalized as a sequence prediction task based on time series data. Recently, more and more deep learning models are used to fit the dynamic stock time series with good nonlinear mapping ability, but not much of them attempt to unveil a market system's internal dynamics. For instance, the driving force (state) behind the stock rise may be the company's good profitability or concept marketing, and it is helpful to judge the future trend of the stock. To address this issue, we regard the explored pattern as an organic component of the hidden mechanism. Considering the effective hidden state discovery ability of the Hidden Markov Model (HMM), we aim to integrate it into the training process of the deep learning model. Specifically, we propose a deep learning framework called Hidden Markov Model-Attentive LSTM (HMM-ALSTM) to model stock time series data, which guides the hidden state learning of deep learning methods via the market's pattern (learned by HMM) that generates time series data. What is more, a large number of experiments on 6 real-world data sets and 13 stock prediction baselines for predicting stock movement and return rate are implemented. Our proposed HMM-ALSTM achieves an average 10% improvement on all data sets compared to the best baseline.

1. Introduction

Predicting the stock status in the future has been long deemed as a significant but challenging task in the research field of financial market analysis. The prediction result of stock movement could serve as a preliminary step for many actual downstream tasks (e.g., stock selection, stock portfolio recommendation and financial distress prediction) (Chang, Tu, Yu, & Qin, 2021; Feuerriegel & Gordon, 2018; Liu, Liu, Zhao, Pan, & Liu, 2020; Qian, Li, & Yuan, 2020; Wang, Cao, Zhao, Liu, & Chen, 2021; Wu, Ma, & Olson, 2022; Zhao, Zhang et al., 2017), so better prediction for stock price provides investors more opportunities to gain profits from the market.

The research for stock price movement prediction mainly contains two primary genres, i.e., fundamental analyses, technical modeling (Xu, Chai, Luo, & Li, 2022). On the fundamental analyses area, researchers focus on evaluating a company's stock by examining its intrinsic value, including but not limited to tangible assets, financial statements, management effectiveness, strategic

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initiatives, and consumer behaviors; essentially all the basics of a company. Hence fundamental analyses usually try to identify internal dynamics of stock markets empirically that determine the long-term trend of stocks. The main idea of technical modeling is to analyze a large amount of market historical data (e.g., time series of stock price and trade volumes), which aims at mining the potential trading pattern to serve for investments in the stock markets. Although technical modeling can better fit the sequence, they usually lack the ability to analyze the future trend from the perspective of the internal driving force of the market, so they are sometimes easily affected by data noise. We mainly focus on technical modeling in this work, and we aim to explore internal dynamics in the technical modeling area.

There has been a variety of inchoate technical works to be proposed in the literature, which range from the early autoregression models to many machine learning algorithms. For the former, autoregression is a statistical method using observations from previous time steps as input to a regression equation to predict the value at the next time step. Some representative models such as AutoRegressive Integrated Moving Average (ARIMA) (Box & Jenkins, 1976) and AutoRegressive Conditional Heteroskedasticity (ARCH) (Engle, 1982) have been used widely in the world (Ariyo, Adewumi, & Ayo, 2014; Dana, 2016; Pai & Lin, 2005). For the machine learning methods, the huge amount of data generated by stock markets forced the researchers to apply machine learning to make investment decisions, where Logistic Regression (Komori et al., 2016), Support Vector Machine (SVM) (Lin, Guo, & Hu, 2013), Hidden Markov Model (HMM) (Gupta & Dhingra, 2012), and other classical machine learning models are applied into the prediction task of future stock markets.

Although the above methods have made considerable progress, the bottleneck caused by the instability and high frequency of stock market time series still makes the results unsatisfactory. Recently, with the rapid development of computing resources and deep learning techniques, neural networks are widely utilized in stock price movement prediction tasks for their stronger fitness caused by better nonlinear mapping ability. Among deep learning models, the RNN (Recurrent Neural Network)-based model Long Short-Term Memory (LSTM) (Hochreiter & Schmidhuber, 1997) neural network dominates in the area of stock prediction because of its sequential-dependent based computational method (Cao, Li and Li, 2019; Chen & Ge, 2019; Du et al., 2021; Liu, Cheng, Su, & Zhu, 2018; Nelson, Pereira, & de Oliveira, 2017; Nguyen, Tran, & Nguyen, 2019; Nikou, Mansourfar, & Bagherzadeh, 2019; Qin et al., 2017; Zhang, Aggarwal, & Qi, 2017).

However, existing deep learning models mainly concentrate on the imitating precision of time series data, which makes the model often confused by the appearance of data for a period leading to wrong judgments. For instance, the driving force (state) behind the similar upward trends of stocks may be the company's good profitability or concept marketing, but the future trends driven by these two states are usually different (Guo, Zou, Zhang, Bo, & Li, 2020). Here the driving state is the so-called internal dynamics. Therefore it is more reasonable for the machine to conclude the pattern from the operation mechanism behind observable time series data. With the learned differentiated states and rules knowledge, the model can make more intelligent and adaptive prediction in an actual forecast. On this point, Wang, Wang, and Wang (2011) proposed the pHMM model to optimize the sequential model from the perspective of system (market) states of different timesteps of the hidden data generating mechanism, and they evaluated the more effective and robust learning mode in the experiments. Unfortunately, the HMM-based model has some disadvantages. First, it lacks sufficient fitting ability when dealing with unstable sequence data such as stock time series; Second, the possible collaborative or competitive relationship between stocks will help to judge the trend of the stock market, while the HMM-based algorithm cannot model multiple stock sequences at the same time. Along this line, considering the powerful non-linear fitting capability of the deep learning model on multivariate series, we aim to integrate the state learning system of HMM into LSTM to promote the efficiency and generalization of existing stock prediction models.

There are three challenges to achieve our goal. First, it is difficult to explore the internal dynamics from time series data and input it into the learned hidden state of LSTM. Second, the LSTM model treats all stock states equally at each moment, failing to distinguish the implicit information between different stocks at each time step. The time point with noise would affect the training process of LSTM. Finally, different from previous internal dynamic learning models based on univariate time series such as pHMM (Wang et al., 2011), our model is crucial to consider that the price of a single stock is often affected by multiple related stocks, even the whole market (Kwon, Choi, & Moon, 2005). A common phenomenon is stock price synchronicity. It indicates that a single stock may be affected by other stocks in the entire market, reflecting the same rise and fall of all stocks in price. For example, stocks under the same industry like GOOGLE (Alphabet Inc.) and FB (Facebook Inc.) might have the same trends in the future. Besides, rising in the stocks of downstream companies tends to promote the stocks of upstream supply companies. Consequently, the objective of this paper is to design the stock prediction model that can adaptively discover the dynamic stock states based on previous stock data. Meanwhile, the learned stock states should reduce the noise in stock time series, and improve the interpretability and robustness compared to general deep learning methods.

To address the challenges mentioned above, in this paper, we propose a novel model called Hidden Markov Model-Attentive LSTM (HMM-ALSTM) and a semi-supervised training algorithm, to explore the daily stock state in the market and estimate the stock price. In order to improve the accuracy of stock movement prediction and stock return prediction, we extract the high-order implicit information through an LSTM-based Encoder-Decoder (Cho et al., 2014) structure. For modeling the stock state, the output of the Decoder is the observation state of the hidden Markov model. We utilize a standard Multi-Layer Perceptron (MLP) to estimate the posterior probability of the stock state. Specifically, stock state modeling helps the model to identify market information from the whole world, which complements the modeling of individual stock local information in stock forecasting. Therefore, even if the price of a single stock is abnormal, the stock status can still reflect the actual value of the stock. For estimating the stock price, the long-term and short-term history information of the stock price will have a certain impact on predicting future stock price. We input the representation learned by the Encoder and Decoder and utilize the Attention mechanism to capture the timing information. Another MLP is implemented to predict the stock movement and stock return rate. We also design maximum likelihood estimation

for the joint distribution as the loss function of stock state modeling. The multi-task learning method is designed to estimate the stock state modeling and stock prediction in parallel. It is proved that adversarial signals are provided among different tasks to improve the robustness and generalization of the model in multi-task learning. In addition, since the importance of different tasks remains uncertain, any subjective weighting is not reliable. Therefore, our weights are generated from the data and trained together with the model as a parameter.

The main contributions of this paper are summarized as follows:

- We propose a novel hybrid structure called HMM-ALSTM to model the stock state and estimate the stock price simultaneously. Further, we develop a maximum likelihood estimation loss function for the training of stock state modeling.
- We utilize the multi-task learning method and trained weight to combine two tasks. We also introduce a Semi-supervised training procedure, which can dynamically train different modules according to the accuracy of different tasks during the training process.
- A large number of experiments have shown that our model performs better than the current state-of-the-art methods for predicting stock movement and return rate. At the same time, we also conduct a detailed analysis of the modeled stock state, which shows its value in guiding investors and analysts on future stock prices.

2. Related work

The related work can be grouped into two categories, i.e., the work on traditional methods and deep learning models for stock prediction.

2.1. Traditional methods

According to the Efficient Market Hypothesis (EMH) (Malkiel, 1989), the history stock time series reveals release of new information, changes in the stock market generally, or random movements around the value that reflects the existing information set. Based on this, stock movement forecasting based on technical modeling intends to predict the short-term trend of stock price time series in the unstable and volatile stock market. Massive traditional methods have been proposed for this, and are mainly divided into two categories, namely traditional time series methods and machine learning methods. Traditional time series methods use historical price data as the basis of estimating future outcomes, among which Arima (Box & Jenkins, 1976) and RACH (Engle, 1982) are two mainstream methods. Further, scholars proposed more variants, e.g. exponentially weighted moving average (Hunter, 1986) and GJR GARCH (Glosten, Jagannathan, & Runkle, 1993). Despite their popularity, traditional time series methods require data to be normally distributed and also that historical data is available to ameliorate prediction. But the stock price time series usually exhibit strong nonlinear trends, seasonal, random walks and are inherently sensitive to a multitude of interdependent factors, which makes traditional time series methods fail.

In contrast with the traditional time series methods, machine learning models make fewer assumptions about the data and have the ability to model nonlinearities, hence gaining prevalence quickly. Classical methods such as SVM (Lin et al., 2013), Random Forest (Patel, Shah, Thakkar, & Kotecha, 2015) and XGBoost (Yun, Yoon, & Won, 2021) have proved the significant improvement in accuracy. As one of the time depending machine learning methods, HMM can model the invisible stock values by its hidden states, which contributes to the extensive applications in the stock prediction area. Recent studies address more importance to the design of mechanisms for HMM. Dias, Vermunt, and Ramos (2015) proposed an extended HMM extracting unobservable heterogeneity between and within stock markets by two designed latent variables. The latter emphasized the application of ensemble learning in stock forecasting. Considering the complex relationships across different markets, Cao, Zhu and Demazeau (2019) developed MCHMM for hierarchical cross-market behavior analysis. They used two coupled HMM layers to explore the relationships within a single market and across multiple stock markets separately. Zhang, Li, Wang, Fang, and Philip (2019) proposed a coupled HMM incorporating the news events with the historical trading data. Experiments showed that HMM could introduce stock related events for stock price forecasting. These works promote the applications of HMM in extracting stock market states. However, their performance in real situations is restricted by two factors. First, the fitting ability relies heavily on feature engineering, which is time consuming and needs expert knowledge. Chaotic and volatile stock data makes feature engineering more difficult, resulting in inaccurate market states modeling. Meanwhile, the stock markets are dynamic and evolutionary, while existing works mainly work on representing global and short-term market states. As the market changes, previous knowledge fails, resulting in a drop in stock prediction accuracy.

2.2. Deep learning models

In recent years, gaining attention has been paid to deep learning methods. The flexibility and the capability to process complexity data outperform other traditional methods in many areas, e.g. image classification (Hu, Peng, Yang, Hospedales, & Verbeek, 2017), text analysis (Stevenson, Mues, & Bravo, 2021), p2p lending (Zhao, Liu et al., 2017), recommendation system (Zhao, Zhao, He, Zhang, & Fan, 2023) and outlier detection (Liu et al., 2019). The use of deep learning methods for stock prediction is also rising. LSTM is a state-of-the-art technique for time series prediction. As an improved version of vanilla RNN, LSTM is introduced to solve the long-term dependency of stock price sequence through a well-designed gate structure. Fischer and Krauss (2018) compared LSTM with random forest, deep neural network and the logistic regression classifier on S&P 500 from 1991 until 2015, to predict

the stock movement. The results showed that the accuracy of LSTM was significantly higher than other models. Jiang, Pan, Jiang, and Long (2018) structured the fundamental influences among homogeneous and heterogeneous markets with three types of correlations, and proposed an improved LSTM called Cd-DLA to learn real-world complex correlations for multiple financial markets prediction. Vo, He, Liu, and Xu (2019) introduced a Deep Responsible Investment Portfolio (DRIP) model that contains a Multivariate Bidirectional LSTM, to predict stock returns for the construction of a socially responsible investment portfolio. Their results proved the performance over other baselines. However, LSTM treats all the hidden states the same, overlooking the difference in importance of the different features.

The advent of the attention mechanism solves the aforementioned problem (Bahdanau, Cho, & Bengio, 2014; Li et al., 2022). It imitates the characteristics of the human brain processing signals and assigns different weights to different signals to stress their importance. In the stock market scenario, the attention mechanism can adaptively learn the importance of data at each time step in the historical data, thereby allocating lower weights to speculative price fluctuations. Qin et al. (2017) proposed an attention-based neural network called DA-RNN to adaptively extract relevant driving series at each time step. The empirical study based on NASDAQ 100 showed that DA-RNN outperformed state-of-art methods for time series prediction. Ding, Wu, Sun, Guo, and Guo (2020) implemented the Multi-Scale Gaussian Prior to enhancing the locality of Transformer in stock movement prediction. Moreover, they also designed a Trading Gap Splitter for Transformer to learn hierarchical features of high-frequency finance data. Experiment results on NASDAQ and Chinese A-share dataset demonstrated the performance of the proposed model. While these studies show the great success of attention-based models in stock prediction, most of the existing methods model the stock state through the stock's historical price sequence but ignore the interaction between the single stock and the entire market.

This work intends to contribute to the literature by introducing a novel hybrid HMM and neural network model called HMM-ALSTM.

3. Data

3.1. Description of data sets

We choose six real word stock indexes on stock market to build our data sets. They cover US stocks and the more volatile Chinese A-shares, thus emphasizing the versatility of HMM-ALSTM. The introduction of the data sets is as follows:

- *CSI (China Securities Index)*: For CSI, we collect three data sets called CSI 100, CSI 300 and CSI 500. The CSI 100 and CSI 200 Indices respectively select the top 100 stocks and the 101st to 300th stocks in the most liquid A-share stocks, reflecting the overall performance of China A-share market. The CSI 500 aims to reflect the overall trend of the stocks of small and medium-sized companies in A-share market. All the three data sets are from 25-JAN-2015 to 27-DEC-2017.
- *GEI (Growth Enterprise Index)*: GEI selects 100 constituent stocks from the ChiNext stocks to reflect the operation of the ChiNext market. We collect data from 25-JAN-2015 to 27-DEC-2017.
- *NASDAQ*: NASDAQ chooses the component stocks from 100 Index, which is a stock market index composed of the 100 largest local and international non-financial listed companies selected based on market capitalization. This data set is from 2-JAN-2019 to 30-DEC-2022.
- *NYSE*: NYSE contains top 100 components of NYSE Composite based on market capitalization. This data set is from 2-JAN-2019 to 30-DEC-2022.

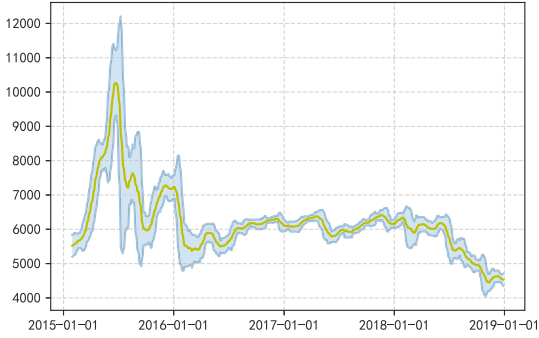
Following Zhang et al. (2017), we set the ground truth as daily-level. Each sample contains 4 features: *open, high, low and close price*. We also introduce 8 technical indicators widely used in technical analysis (Kirkpatrick & Dahlquist, 2010), as shown in Table 1. All the technical indicators are calculated by the Python package TA-Lib. For brevity, the input features for every timestep contain 12 features overall. Different stocks have different opening prices, closing prices, highest prices, and lowest prices, and there is a large gap between the mean and variance of these prices. This is not conducive to the convergence of the trained model. Therefore, it is necessary to normalize these input features before feeding them into our model. It is unreasonable to directly normalize time-series data over time, since earlier data uses future information. Therefore, we adopt a normalized rolling method similar to the training method, for each sliding window, we calculate the mean and variance of each feature of the input data within the sliding window, and use this to normalize the original features. For each stock, the first 80% days are used to construct the training set and the last 20% are used as test set. This method better resembles the forecasting scenarios in practice and thus allows us to conduct a real assessment of the forecasting performance.

3.2. Volatility between stock markets

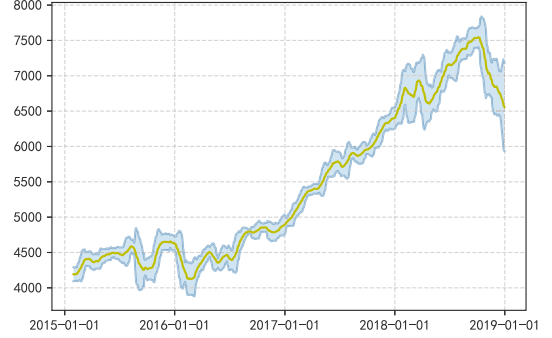
In this section, we further analyze the volatility of different stock market data. Specifically, we use Bollinger Bands to visualize the data of the CSI 100 and NASDAQ 100 indexes from 2015 to 2018. Bollinger Bands are a type of statistical chart characterizing the prices and volatility over time of a financial instrument or commodity. Bollinger Bands utilize the stock price channel to display the volatility. When the stock price fluctuates very little and is in consolidation, the stock price channel will become narrow, which may indicate that the stock price fluctuation is in a temporary calm period. Conversely, it implies that the stock price is about to produce violent fluctuations. As Fig. 1 shows, compared with NASDAQ 100, the stock price channel of CSI 100 is wider, indicating that the volatility of its constituent stocks is stronger, which was particularly obvious in 2015.

Table 1
Technical indicators.

Indicator	Category
Awesome oscillator	Momentum
Money flow index	Momentum
Chaikin money flow	Volume
On-balance volume mean	Volume
Bollinger Bands (Upper)	Volatility
Bollinger Bands (Lower)	Volatility
Average directional movement index	Trend
Moving average convergence divergence	Trend



(a) Bollinger Bands on CSI 100



(b) Bollinger Bands on NASDAQ 100

Fig. 1. Volatility comparison between A shares and US stocks.

4. Models

In this section, we first introduce some preliminary knowledge of HMM and LSTM. Then, we give a specific definition of the research problem and present details of the three modules in our proposed HMM-ALSTM. Further, we propose a semi-supervised training procedure utilizing multi-task learning method for our model. Important notations of HMM-ALSTM are outlined in Table 2.

4.1. Preliminaries

Here we present some preliminary knowledge of HMM and LSTM.

HMM models the real observed data and implicit information by two different states called the observation state and the hidden state separately. The hidden state s_t at time t is unobservable and produced by a stochastic process. And with the first-order Markov property, the hidden state s_t only depends on s_{t-1} with specific state transition probability.

In this paper, to characterize an HMM, four elements are necessary, including the number of hidden states N , the state transition probability, the emission probability and the initial state distribution Π . N is equal to the number of stock states. The state transition probability and the emission probability are denoted as $P(s_t|s_{t-1})$ and $P(o_t|s_t)$ at time t . And the transition probability is denoted as $\Pi = \{\pi_i | 1 \leq i \leq N\}$, which means the probability of each hidden state appearing at time 0. For the stock state modeling task, we utilize the hidden state s_t to characterize the stock state at time t . More specifically, we need to find the most possible stock state sequence S_T , given the observation sequence O_T and the four elements. This is an unsupervised learning problem because the stock state is not predefined, which is called decoding problem and solved by the Viterbi Algorithm (Viterbi, 1967). It defines two local states based on dynamic programming. The first one is the maximum probability among all possible state transition paths where the hidden state S_{t-1} is i at time t , and denoted as $\delta_t(i)$,

$$\begin{aligned} \delta_t(i) &= \max_{X_t, S_{t-1}} P(s_t = i, S_{t-1}, O_t) \\ &= \max_{1 \leq j \leq N} [\delta_{t-1}(j) P(s_t = i | s_{t-1} = j)] P(o_t | s_t = i) \end{aligned} \quad (1)$$

The second local state is recursively obtained from the first local state. It is defined as the hidden state at time $t-1$ of the most possible state transition path, and is denoted as $\Psi_t(i)$,

$$\Psi_t(i) = \arg \max_{1 \leq j \leq N} [\delta_{t-1}(j) P(s_t = i | s_{t-1} = j)] \quad (2)$$

With these two local states, we can get the most possible S_T through backtracking $\Psi_t(i)$.

Table 2
Important notations.

Notations	Descriptions
X_T	The sequence of features.
P_T	The sequence of stock price features.
Q_T	The sequence of technical indicator features.
S_T	The stock state sequence learned by HMM.
$y_i^{M,n}$	The ground truth of n -day stock movement at time i .
$y_i^{R,n}$	The ground truth of n -day return rate at time i .
$\hat{y}_i^{M,n}$	The predicted n -day stock movement at time i .
$\hat{y}_i^{R,n}$	The predicted n -day return rate at time i .
h_i^e	The hidden state of LSTM learned by Encoder at time i .
h_i^d	The hidden state of LSTM by Dncoder at time i .
c_i^e	The cell state of LSTM learned by Encoder at time i .
c_i^d	The cell state of LSTM by Dncoder at time i .
β_i	The attention score at time i .
x^{att}	The aggregated feature.
O^T	The observable feature sequence learned by Decoder.
α_T	The joint distribution of sequence.

LSTM (Hochreiter & Schmidhuber, 1997) was proposed to address the vanishing gradients problem in vanilla RNN, through adding cell states to track long-term dependency. The input data for LSTM is denoted as $X_T = \{x_t | t = 1, 2, 3, \dots, T\}$. For every timestep, $x_t \in \mathbb{R}^D$ is an input vector with D dimensions. Vectors c^t and $h^t \in \mathbb{R}^U$ denote the cell state vector and the hidden state vector respectively, where U is the number of hidden units. Vector $z^t \in \mathbb{R}^U$ is an information transformation module. Vector i^t, o^t and $f^t \in \mathbb{R}^U$ denote the input, output, and forget gate vector. Following equations define the whole process:

$$\begin{aligned}
 z^t &= \tanh(W_z x^t + Q_z h^{t-1} + b_z) \\
 i^t &= \sigma(W_i x^t + Q_i h^{t-1} + b_i) \\
 f^t &= \sigma(W_f x^t + Q_f h^{t-1} + b_f) \\
 c^t &= f^t \odot c^{t-1} + i^t \odot z^t \\
 o^t &= \sigma(W_o x^t + W_h h^{t-1} + b_o) \\
 h^t &= o^t \odot \tanh(c^t)
 \end{aligned} \tag{3}$$

where $W_z, W_i, W_f, W_o \in \mathbb{R}^{U \times D}$, and $Q_z, Q_i, Q_f \in \mathbb{R}^{U \times U}$ are mapping matrices; $b_z, b_i, b_f, b_o \in \mathbb{R}^{U \times U}$ are bias vectors.

Hidden states are introduced in both HMM and LSTM, and they model the implicit information of observations data. The difference is that the hidden state of HMM obeys the first-order Markov assumption, which limits the fitting ability of HMM in complex systems. At the same time, the hidden state of LSTM lacks probabilistic relations, resulting in a lack of interpretability of the model. At the same time, targeting only fitting time series observations can be misleading by noisy information, especially in the stock market. Therefore, in the stock forecasting task, HMMs and LSTMs face different challenges. For HMM, limited by the ability of feature extraction, it can only learn potential signals in stock prices under strict assumptions about the data. This greatly weakens the performance of HMM on the stock prediction task. On the contrary, the results of LSTM learning lack interpretability, which makes it impossible to explain the reason when the model predicts large fluctuations. Therefore, our proposed HMM-ALSTM eliminates these defects through the combination of the two models.

4.2. Problem formalization

Consider a time series of stock price $P_T = \{p_t | t = 1, 2, 3, \dots, T\}$, we denote the sequence of open, high, low, and close price as $p_t = [p_t^o, p_t^h, p_t^l, p_t^c]$. The historical price sequence can be simplified as $P_{t-n,t} = \{p_{t-n}, p_{t-n+1}, \dots, p_t\}$, where n is the window length. Meanwhile, we denote m technical indicator vector at time t as $q_t = [q_t^1, q_t^2, \dots, q_t^m]$, where q_t^i is related to historical price sequence with $q_t^i = f(P_{t-n,t}; \theta^i)$, and θ^i is the parameter of technical strategy i . Similarly the technical indicator sequence is represented as $Q_T = \{q_t | t = 1, 2, 3, \dots, T\}$. The two sequences P_T and Q_T are concated to get the input features $X_T = \{x_t | t = 1, 2, 3, \dots, T\}$. We describe the stock prediction problem as Task 1, and Task 2 is able to help the model to be more robust and improve on task 1:

(Task 1). Stock prediction Stock prediction is divided into two challenging tasks, namely stock movement prediction and stock return prediction. Given the price sequence P_T and technical indicator sequence Q_T , the stock movement prediction can be regarded as a binary classification task. The n -day stock movement after time t is defined as

$$y_i^{M,n} = \begin{cases} 1, & p_t^c > p_{t-n}^c \\ 0, & p_t^c \leq p_{t-n}^c \end{cases} \tag{4}$$

For the stock return problem, the goal of our model is to predict the n -day return ratio of a stock, which is defined as

$$y_i^{R,n} = \frac{p_t^c - p_{t-n}^c}{p_{t-n}^c} \tag{5}$$

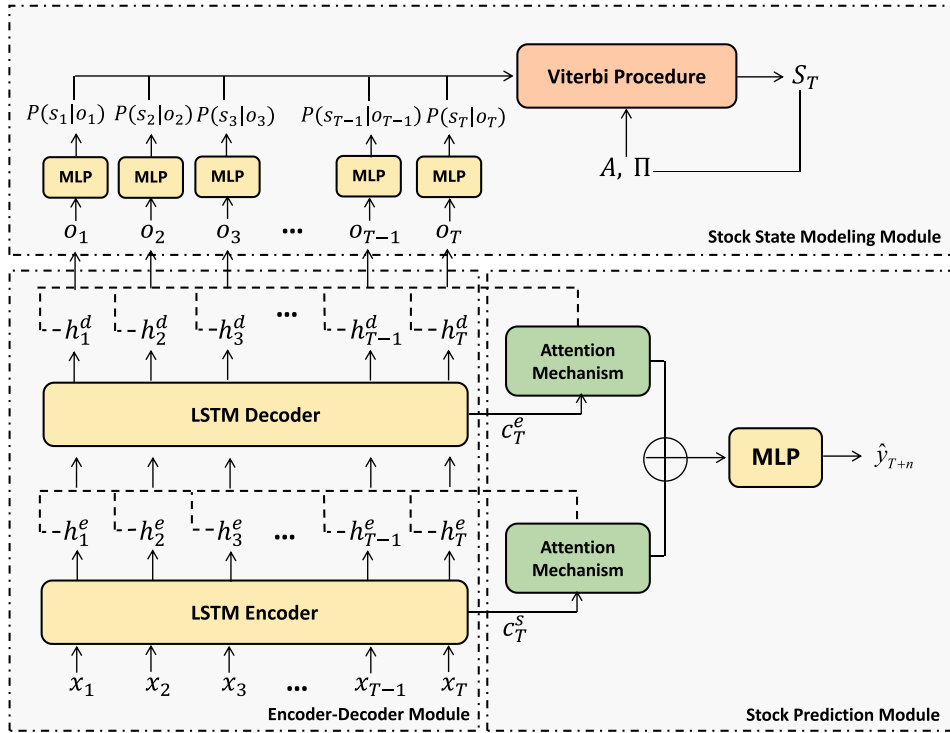


Fig. 2. Overall structure of HMM-ALSTM.

Task 2. Stock state modeling According to Caruana (1997), domain-specific information contained in the training signals of related auxiliary tasks can improve the performance of main task. So we introduce the stock state modeling task as the auxiliary task. Specifically, we utilize the hidden states of HMM to characterize the stock states. The whole task is divided into two processes, one of them is to learn the stock state that reflects market-level internal dynamics at each time step based on the observation states. In our work we use neural network to learn high-order features from raw price series, and these features are the observation states. Based on the decoding problem mentioned in Section 4.1, this process can be solved by estimating the most possible stock states $S_T = \{s_t | t = 1, 2, 3, \dots, T\}$. Another process is to train the neural network based on the learned stock states, which can be regarded as a multi-classification task at each time step if $N > 2$. This process is the auxiliary task of task 1 to integrate the information of internal dynamics.

4.3. Overall structure of HMM-ALSTM

We introduce the structure of our framework in detail in this section. Fig. 2 shows the overall structure of the proposed model. Corresponding to the three panels in this figure, to cope with the stock prediction target and stock state modeling at the same time, our proposed model is divided into three modules. First, we utilize an LSTM-based Encoder–Decoder structure to capture, transform and aggregate the sequential information in the input sequence. The second part takes the hidden states of Decoder as the observation states of HMM. Specifically, the HMM uses the high-order features extracted by the neural network as the observation state, and uses the Viterbi algorithm to adaptively learn the model parameters of the HMM. During training, the fitting effect of HMM will be fed back to the Encoder–Decoder structure as a loss function. Thirdly, the stock prediction module is designed with two global attention mechanisms for the long-time dependence of both Encoder and Decoder. And the output of the attention mechanisms is concatenated and processed by an Multi-Layer Perceptron (MLP). The final predicted results $\hat{y}_{T+n}$ for the stock prediction problem can be obtained by the MLP. Moreover, as there are commonalities between the two tasks, we utilize multi-task learning to predict the state sequence and ground truth at the same time.

4.4. Encoder–Decoder module

The Encoder–Decoder module follows the classic Encoder–Decoder structure widely used in machine translation (Cho et al., 2014). Specifically, the Encoder–Decoder structure is composed of two LSTMs, and the output of one LSTM is used as the input of

the other LSTM. The Encoder extracts the shallow semantic information in the historical price of the stock, and the input of each time step x_t contains the price and technical indicators. We utilize $LSTM^e$ to represent the single cell mentioned in Eq. (3),

$$\begin{aligned} h_t^e &= LSTM(h_{t-1}^e, x_t) \\ h_T^e, c_T^e &= LSTM(h_{T-1}^e, x_T) \end{aligned} \quad (6)$$

where h_t^e is the hidden state for Encoder at time t . c_T^e represents the output cell state for Encoder at the last time step.

The outputs of Encoder are the hidden state sequence $\mathbf{h}^e = \{h_t^e | t = 1, 2, 3, \dots, T\}$ and the cell state c_T^e . Unlike text data where each token is affected by the context, stock price series is not affected by future prices. To prevent data leakage, the Decoder only takes $\mathbf{h}^e$ the encoder as the input sequence to extract high-order features that characterize the state of the stock,

$$\begin{aligned} h_t^d &= LSTM(h_{t-1}^d, h_t^e) \\ h_T^d, c_T^d &= LSTM(h_{T-1}^d, h_T^e) \end{aligned} \quad (7)$$

where h_t^d is the hidden state for Decoder at time t . c_T^d represents the cell state of Decoder. The Decoder outputs the hidden states sequence $\mathbf{h}^d = \{h_t^d | t = 1, 2, 3, \dots, T\}$ and global information c_T^d . Different from the Encoder, the Decoder takes the high-order features learned by the Encoder as input instead of the original feature x_t . At the same time, the output of the Decoder serves the HMM as the observation states.

4.5. Stock prediction module

The stock prediction modules combine the price dynamics information features from Encoder and the high-order stock states information from Decoder. Intuitively, there are certain patterns in historical data that are helpful for predicting future stock price and movement. However, these patterns are easily drowned out in the noise generated by investor sentiment and market distortions. Therefore, we implement the attention mechanism to capture the dynamic temporal correlation indirectly (for example, intermittent rise and fall of stock price during structural adjustment), so as to adaptively obtain effective sequence patterns. More specifically, we compute the attention value for each hidden state with cell state, and weighted sum all hidden states, as follows Eqs. (8), (9), (10).

$$Score(h_t, c_T) = v^T \tanh(W_1 h_t + W_2 c_T) \quad (8)$$

$$\beta_t = \frac{\exp(Score(h_t, c_T))}{\sum_{i=1}^T \exp(Score(h_i, c_T))} \quad (9)$$

$$x^{att} = \sum_T \beta_t h_t \quad (10)$$

where W_1 , W_2 and v^T are learnable parameters. Then the MLP receives the temporal attention vector x^{att} from encoder and decoder and output the results. For stock return prediction, as it is a regression problem, we utilize the mean squared error as Eq. (11) for optimization,

$$Loss_1^R = \frac{1}{n} \sum_{i=1}^{|Z|} (y_i^R - \hat{y}_i^R)^2 \quad (11)$$

where y_i^R is the ground truth of return rate and $\hat{y}_i^R$ is the predicted value.

For stock movement prediction, the binary cross entropy loss as Eq. (12) is utilized for training.

$$Loss_1^M = -\frac{1}{n} \sum_{i=1}^{|Z|} [y_i^M \log(\hat{y}_i^M) + (1 - y_i^M) \log(1 - \hat{y}_i^M)] \quad (12)$$

where y_i^M is the target for movement prediction and $\hat{y}_i^M$ is the prediction.

4.6. Stock state modeling module

The stock state modeling module takes the output of Decoder as the observation sequence $O_T = \{o_t | t = 1, 2, 3, \dots, T\}$. Following Section 4.2, given the O_T and the number of stock state N , the aim is to train a HMM that the output of HMM should be the most probable state sequence as Eq. (13).

$$\hat{S}_T = \underset{S_T}{\operatorname{argmax}} P(S_T | O_T) \quad (13)$$

As the definition of HMM, the joint distribution of the partial observation and state sequence until time t can be denoted as Eq. (14).

$$\begin{aligned} \alpha_t &= P(S_t, O_t) = P(s_t, o_t, S_{t-1}, O_{t-1}) \\ &= P(s_t, o_t | S_{t-1}, O_{t-1}) P(S_{t-1}, O_{t-1}) \\ &= P(o_t | s_t, S_{t-1}, O_{t-1}) P(s_t | S_{t-1}, O_{t-1}) \alpha_{t-1} \end{aligned} \quad (14)$$

where $P(s_t | S_{t-1}, O_{t-1})$ and $P(o_t | S_{t-1}, O_{t-1})$ are called state transition probability and emission probability in HMM. State transition probability only depends on the state of the last timestep for one-order HMM and is equivalent to $P(s_t | s_{t-1})$. And emission probability

is assumed dependent on the current state only, which is equivalent to $P(o_t|s_t)$. It has been proved that output of a standard MLP with the softmax layer for classification can be interpreted as estimates for posteriori probability. Thus, we utilize a MLP to estimate the posteriori probability $P(s_t|o_t)$. Since the number of stock state is N , the output of softmax layer is the posteriori possibility of every stock state. And this process can be seen as a multi-classification problem if $N > 2$. The emission probability can be estimated based on Bayes formula as follows.

$$P(o_t|s_t) = \frac{P(s_t)}{P(s_t|o_t)P(o_t)} \quad (15)$$

where $P(o_t)$ is the prior probability for o_t and is a constant. As for $P(s_t)$, it can be calculated by the data set. And given the initial state distribution $\Pi = [\pi_i|i = 1, 2, 3, \dots, N]$, the initialization and termination for α_t are as follows based on Eq. (14),

$$\alpha_t = \begin{cases} \pi_{s_1} P(o_1|s_1), & t = 1 \\ P(o_t|s_t)P(s_t|s_{t-1})\alpha_{t-1}, & t > 1 \end{cases} \quad (16)$$

Considering the time dependence of the stock state, we utilize maximum likelihood estimation for the joint distribution α_T as the loss function to train the parameters. The aim of maximum likelihood training is to simultaneously maximize the log probability of all the observation sequences and state sequences. Given a training set Z , the minimizing formation is defined as Eq. (17),

$$Loss_2 = - \sum_{(X,S) \in Z} \ln[P(S|O)] = - \sum_{(X,S) \in Z} \ln[(\alpha_T / P(O))] \quad (17)$$

where $P(O)$ is a constant during training, thus $Loss_2$ can be simplified as follows,

$$Loss_2 = - \sum_{(X,S) \in Z} \ln[(\alpha_T)] \quad (18)$$

For estimating the most possible state sequence, it is the same as the decoding problem of HMM and can be solved by the Viterbi algorithm mentioned in Section 4.2.

4.7. Semi-supervised training procedure

We integrate the loss function $Loss_1$ for stock prediction and $Loss_2$ for stock state modeling to get the multi-task learning formation. Traditional methods utilize a weighted linear sum of different loss functions, but the subjectively defined weights bond the performance. Thus we utilize a multi-task loss function with homoscedastic uncertainty (Kendall, Gal, & Cipolla, 2018) as Eq. (19). Specifically, this loss function assumes likelihood as a Gaussian with mean given by the model output. Consequently, we interpret minimizing this last objective with respect σ as learning the relative weight of the losses adaptively. As the noise decreases, we have that the weight of the respective objective increases. When the noise rises, vice versa.

$$Loss_{total} = \sum_{i=1}^2 \frac{1}{2\sigma_i^2} Loss_i + \log \sigma_i \quad (19)$$

To train the NN-HMM, we propose a semi-supervised training procedure to make our model self-learn meaningful stock state and the ground truth.

Algorithm 1: Semi-supervised training procedure

Input: Feature sequence: X ; Stock prediction ground truth: Y ; Pre-clustered state sequence: S_{pre} ; Threshold for state transition matrix similarity: γ

Output: Predicted ground truth: $\hat{Y}$; State sequence: $\hat{S}$.

- 1 Initialize model parameters Θ randomly;
- 2 Calculate the state transition matrix A and initial state distribution Π by S_{pre} ;
- 3 **while** state transition matrix similarity $d > \gamma$ **do**
- 4 **while not converged do**
- 5 Calculate $\hat{Y}$ and $\hat{S}$;
- 6 Update Θ by Eq. (19);
- 7 **end**
- 8 Update A and Π by $\hat{S}$;
- 9 Update $\hat{S}$ by viterbi procedure;
- 10 Calculate $\hat{A}$ and $\hat{\Pi}$;
- 11 Calculate the similarity between A and $\hat{A}$;
- 12 Update A and Π by $\hat{A}$ and $\hat{\Pi}$;
- 13 **end**

As shown in Algorithm 1, the training process needs an initial state as the input to train our proposed neural network. Therefore, we sampled from the raw data, trained a k-means clustering, and calculated the stock states for the rest of the training samples using

the well-trained clustering model. With the initial stock states and ground truth, the parameters of the neural network part can be updated by gradient descent. Meanwhile, the HMM updates state transition matrix $\hat{A}$ and Π through by counting frequency. Each subsequent iteration is the same as the first iteration, except for updating stock state sequence by Viterbi Algorithm. We define the state transition similarity as Eq. (20).

$$\delta = \sum_{i=1}^N \frac{a_i \hat{a}_i}{\|a_i\| \|\hat{a}_i\|} \quad (20)$$

where a_i and $\hat{a}_i$ are the row vector of A and $\hat{A}$, and the iterations will stop until δ is greater than the threshold we defined.

5. Experiments

In this section, we first introduce our experimental setup in detailed. Then, a large number of experiments are conducted to show the performance of HMM-ALSTM compared of the current state-of-the-art methods for stock movement and return ration prediction.

5.1. Experimental setup

Compassion Methods. We illustrate the performance of HMM-ALSTM with other baseline methods over the six real word data sets.

- **AR** (Autoregressive model) is a classical method that is widely used for time-series forecasting. It predicts the future market price from the prices of several past steps.
- **LR** (Linear Regression & Logistic Regression) is the linear approach for modeling the relationship among the features. We use logistic regression and linear regression in stock movement prediction and stock return prediction, respectively.
- **ARIMA** (Autoregressive Integrated Moving Average) is one of the traditional time series model combining differencing with the autoregressive model and the moving average model (Box & Jenkins, 1976).
- **GARCH** (Generalized Autoregressive Conditional Heteroskedasticity Model) is a traditional time series model used to describe a changing, possibly volatile variance (Engle, 1982).
- **LightGBM** (Light Gradient Boosting Machine) is a gradient boosting framework that uses tree based learning algorithms (Ke et al., 2017).
- **TextCNN** is a convolutional neural network architecture applied to sentence classification, it extracts short-term patterns from the price sequence (Kim, 2014).
- **RNN** (Recurrent Neural Network) is the basic RNN utilizing internal state (memory) to process variable length sequences of inputs (Rumelhart, Hinton, & Williams, 1986).
- **LSTM** (Long Short-Term Memory networks) is a recurrent neural network architecture that can avoid vanishing gradient, which is a useful tool for modeling sequence (Hochreiter & Schmidhuber, 1997).
- **TCN** (Temporal Convolutional Network) incorporates the causal convolutions and dilated convolutions to enhance the ability in capturing long-term dependency (Bai, Kolter, & Koltun, 2018).
- **ALSTM** (Attention LSTM) adds an external attention layer into the LSTM architecture to attentively aggregate information from all hidden states in previous timestamps (Qin et al., 2017).
- **SFM** (State Frequency Memory recurrent network) is a redesigned recurrent neural network by decomposing the hidden states of LSTM into multiple frequency components to model multi-frequency trading patterns (Zhang et al., 2017).
- **ADARNN** (Adaptive RNNs) proposes Temporal Distribution Characterization and Temporal Distribution Matching to tackle the distribution shift problem statistical properties change over time (Du et al., 2021).
- **CATN** develops an end-to-end model of Cross Attentive Tree-aware Network to jointly capture the inter-series correlation and intra-series temporal patterns (He, Zhang, Bai, Yi, & Niu, 2022).

Evaluation Metrics. To evaluate the forecasting performances, here we select three widely used metrics for stock movement prediction, i.e., the *Accuracy Rate*, *Precision Rate* and *Recall Rate*. For return rate prediction, two popular metrics for regression problems are selected, i.e., the *Mean Absolute Error (MAE)* and *RootMean Squared Error (RMSE)*.

Implementation details. The proposed HMM-ALSTM is implemented in Pytorch, and the dimensions of the hidden state for LSTM Encoder and Decoder are 64. For stock state modeling, we utilize a two-layer MLP with hidden size 128 and 128. And a three-layer with hidden sizes 256, 128, and 128 are used for stock prediction. The hidden state number is 6 for the best performance. For the baselines, the hidden size for RNN and LSTM is set to be 128. For LightGBM, the number of base estimators is 1000 and the learning rate is set to be 0.05. In addition, the max depth for each base model is 5. And for other baselines, we follow the best parameter setting reported in the original papers. For all the gradient descent-based methods, we use the Adam optimizer with the initial learning rate of 0.0005 and the batch size of 64 to optimize the model.

Table 3

Performance over baselines on 1-day movement prediction of China A shares.

Model	CSI 100			CSI 200			CSI 500			GEI 100		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
AR	0.5602	0.6214	0.6405	0.5437	0.6071	0.6549	0.5231	0.5981	0.6114	0.5836	0.6432	0.6512
LR	0.5933	0.6544	0.6633	0.5463	0.6525	0.6329	0.5518	0.6703	0.6481	0.5977	0.6765	0.6528
ARIMA	0.6015	0.6771	0.6856	0.5714	0.6672	0.6703	0.5747	0.6792	0.6914	0.6152	0.6772	0.6664
GRACH	0.6024	0.6758	0.6911	0.5759	0.6644	0.6815	0.5789	0.6808	0.6962	0.6121	0.6803	0.6689
LightGBM	0.6235	0.6911	0.6993	0.6049	0.6822	0.7003	0.5927	0.6886	0.7062	0.6322	0.7341	0.6912
TextCNN	0.6314	0.6902	0.7345	0.6108	0.7001	0.6985	0.6115	0.7075	0.6753	0.6273	0.7346	0.6809
RNN	0.6143	0.7039	0.7150	0.6028	0.7103	0.6947	0.5879	0.7161	0.7097	0.6313	0.7157	0.7230
LSTM	0.6305	0.7113	0.7393	0.6251	0.7106	0.7006	0.5967	0.7146	0.7142	0.6438	0.7237	0.7152
TCN	0.6454	0.7306	0.7526	0.6414	0.7492	0.7323	0.6192	0.7316	0.7522	0.6681	0.7654	0.728
ALSTM	0.6447	0.7292	0.7522	0.6351	0.7524	0.7204	0.6245	0.7317	0.7541	0.6679	0.7750	0.7308
SFM	0.6547	0.7314	0.7565	0.6474	0.7442	0.7347	0.6215	0.7373	0.7547	0.6592	0.7603	0.7276
ADARNN	0.6458	0.7267	0.7441	0.6402	0.7503	0.7352	0.6122	0.7293	0.7485	0.6701	0.7819	0.7254
CATN	0.6511	0.7313	0.7531	0.6431	0.7481	0.7335	0.6118	0.7289	0.7493	0.6635	0.7716	0.728
HMM-ALSTM	0.6622	0.7461	0.7687	0.6591	0.7574	0.7433	0.6262	0.7337	0.7594	0.6722	0.7761	0.7379
Max	0.1020	0.1247	0.1282	0.1154	0.1503	0.1104	0.1031	0.1356	0.1480	0.0886	0.1329	0.0867
Min	0.0075	0.0147	0.0122	0.0117	0.0050	0.0081	0.0017	−0.0036	0.0047	0.0021	−0.0058	0.0071

Table 4

Performance over baselines on 1-day return prediction of China A shares.

	CSI 100		CSI 200		CSI 500		GEI 100	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
AR	0.1033	0.1362	0.1152	0.1563	0.1295	0.1617	0.0961	0.1123
LR	0.0838	0.1053	0.1027	0.1385	0.1144	0.1323	0.0742	0.0972
ARIMA	0.0775	0.0965	0.0944	0.1122	0.1056	0.1286	0.0621	0.0813
GRACH	0.0752	0.0972	0.0965	0.1165	0.1024	0.1247	0.0667	0.0896
LightGBM	0.0662	0.0876	0.0744	0.0963	0.0885	0.1194	0.0434	0.0733
TextCNN	0.0544	0.0744	0.0578	0.0852	0.0661	0.0831	0.0553	0.0799
RNN	0.0634	0.0841	0.0631	0.0769	0.0772	0.0985	0.0629	0.0844
LSTM	0.0459	0.0725	0.0466	0.0713	0.0674	0.0849	0.0408	0.0652
TCN	0.0275	0.0478	0.0288	0.0503	0.0405	0.0592	0.0261	0.0558
ALSTM	0.0268	0.0469	0.033	0.0544	0.0431	0.0589	0.023	0.0537
SFM	0.0247	0.0445	0.0269	0.0495	0.0392	0.0594	0.0292	0.0604
ADARNN	0.0234	0.0421	0.0223	0.0452	0.0264	0.0466	0.0264	0.0582
CATN	0.0256	0.0465	0.0274	0.0499	0.0315	0.0513	0.0288	0.0576
HMM-ALSTM	0.0192	0.0338	0.0209	0.0405	0.0295	0.0485	0.0184	0.0327
MAX	0.0841	0.1024	0.0943	0.1158	0.1000	0.1132	0.0777	0.0796
MIN	0.0042	0.0083	0.0014	0.0047	−0.0031	−0.0019	0.0046	0.0210

5.2. Main results

Tables 3–6 demonstrate the performance of 1-day stock prediction task on China A shares and US stocks. We first note that all the implemented deep learning models show noticeably better performance compared to other traditional methods. All deep learning models achieve at least 5% and 7% improvement in the accuracy of stock movement prediction and MAE of stock return prediction respectively. This difference in performance can be attributed to the considerably stronger nonlinear mapping ability of deep learning methods. The exception to this is LightGBM where, for all the data sets, the performance of LightGBM reaches or even outperforms the classic RNN and TextCNN. As an ensemble model, LightGBM is an integration of classic decision trees, which itself is an effective nonlinear mapping method. This contributes to the ability of LightGBM to process stock time series. Meanwhile, there is a lot of randomness, noise and ambiguity in the stock market, which can mislead the deep learning model overfitting during training. The ensemble model introduces variance through the combination of multiple weak base models, thereby effectively alleviating overfitting.

Considering deep learning methods, the proposed HMM-ALSTM is almost superior to all baseline methods in any metrics. Specifically, HMM-ALSTM improves over the strongest baselines with 0.0069, 0.0080, and 0.0066 on average Accuracy, Precision and Recall score of stock movement, and 0.0031, 0.0054 on average MAE and RMSE of return rate. In addition, deep learning models with attention mechanisms such as ALSTM and the proposed HMM-ALSTM achieve significant improvements, which can adaptively mine the important signals in the stock sequence for predicting future price movements. And we can easily see the proposed HMM-ALSTM shows higher accuracy and robustness over ALSTM, SFM, and ADARNN. This is perhaps because these three models only focus on improving the performance of a single stock sequence but overlook the influences related to stocks in the same industry, which implicitly and comprehensively reflects the overall trends of the stock market. This is in line with the explanation of stock price co-movement in behavioral finance theory. Correspondingly, the proposed HMM-ALSTM captures the market-level relevance through the stock state modeling module, resulting in improvements in prediction performance. The exception to this is

Table 5
Performance over baselines on 1-day movement prediction of US stocks.

Model	NASDAQ 100			NYSE 100		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall
AR	0.5916	0.6813	0.6515	0.5812	0.6135	0.6359
LR	0.615	0.716	0.6723	0.5903	0.6581	0.654
ARIMA	0.6238	0.7211	0.6848	0.5985	0.6711	0.6628
GRACH	0.6284	0.7272	0.6821	0.6005	0.6758	0.6821
LightGBM	0.6409	0.746	0.7298	0.61	0.7066	0.7117
TextCNN	0.6532	0.7527	0.718	0.6211	0.7135	0.7064
RNN	0.6451	0.7552	0.734	0.619	0.7085	0.71
LSTM	0.651	0.7652	0.7216	0.6308	0.7194	0.7029
TCN	0.6604	0.76	0.7431	0.6568	0.7255	0.7102
ALSTM	0.6609	0.7524	0.7399	0.6549	0.7344	0.725
SFM	0.6625	0.7615	0.7334	0.6589	0.7229	0.7097
ADARNN	0.6712	0.7666	0.759	0.632	0.7096	0.722
CATN	0.6679	0.7671	0.7432	0.6408	0.7202	0.7169
HMM-LSTM	0.6784	0.7775	0.7571	0.6618	0.7521	0.7282
Max	0.0868	0.0962	0.1056	0.0806	0.1386	0.0923
Min	0.0072	0.0109	−0.0019	0.0029	0.0177	0.0032

Table 6
Performance over baselines on 1-day return prediction of US stocks.

Model	NASDAQ 100		NYSE 100	
	MAE	RMSE	MAE	RMSE
AR	0.0962	0.1303	0.1135	0.1454
LR	0.0788	0.0112	0.0815	0.1033
ARIMA	0.0615	0.0821	0.0744	0.0889
GRACH	0.0564	0.0743	0.0642	0.0832
LightGBM	0.0413	0.0652	0.0612	0.0801
TextCNN	0.0385	0.0583	0.0482	0.0657
RNN	0.0422	0.0642	0.0558	0.0796
LSTM	0.0339	0.0515	0.0436	0.0598
TCN	0.0191	0.0366	0.0226	0.0451
ALSTM	0.0182	0.0341	0.0211	0.0432
SFM	0.0198	0.0376	0.0227	0.0453
ADARNN	0.0162	0.0258	0.0189	0.0386
CATN	0.0172	0.0275	0.0206	0.0405
HMM-LSTM	0.0101	0.0216	0.0142	0.0275
Max	0.0861	0.1087	0.0993	0.1179
Min	0.0061	0.0042	0.0047	0.0111

for return prediction on CSI 500, the performance of all baselines drops and, the ADARNN outperforms the proposed HMM-ALSTM. This could be attributed to the greater volatility and uncertainty in the stock price of small and medium-sized companies in A shares, and the rise and fall of their stock price are more susceptible to their own operating conditions rather than the stock market. This means that the stock state modeling task actually introduces new noise to the training of the stock prediction task, which leads to a drop in prediction performance. Across both stock markets, we can see that the fitting degree of HMM-ALSTM on A shares is worse than that of US stocks, reflected in all the baselines. This is related to the higher volatility of the A shares we analyzed in Section 3. In fact, the deep learning methods, especially the proposed HMM-ALSTM, have a better effect on the unstable market fitting, and the performance decline is relatively small compared with the traditional methods. This shows the superiority of the nonlinear mapping ability of the deep models for unstable time series data.

We also show the performance of 3-day and 7-day stock prediction in supplementary material. For 3-day stock prediction, HMM-ALSTM improves over the strongest baselines with 0.0064, 0.0073, and 0.0063 on average Accuracy, Precision and Recall score of stock movement, and 0.0029, 0.0047 on average MAE and RMSE of return rate. For 7-day stock prediction, HMM-ALSTM improves over the strongest baselines with 0.0067, 0.0079, and 0.0072 on average Accuracy, Precision and Recall score of stock movement, and 0.0031, 0.0044 on average MAE and RMSE of return rate. These results prove the accuracy and robustness.

From these results, we can conclude that the deep learning methods do show a significant performance improvement than traditional methods such as ARIMA and GRACH, which supports that deep learning methods can better mine information from noisy and chaotic stock time series. Based on the strengths of the deep learning methods, the stock modeling task adopted by HMM-ALSTM can effectively capture the dynamic state information in the complex stock market. And fusing this information with stock price data can significantly improve the performance of stock prediction tasks, especially when compared to other deep learning baselines based on attention mechanisms. Our findings thus far only consider the overall performance of HMM-ALSTM on the test set.

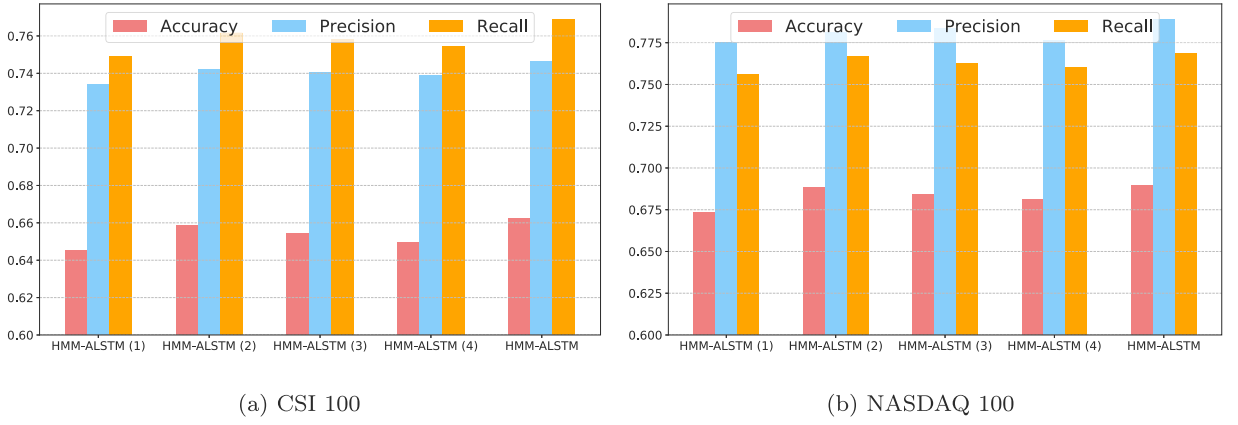


Fig. 3. The performance of HMM-ALSTM on 1-day stock movement prediction.

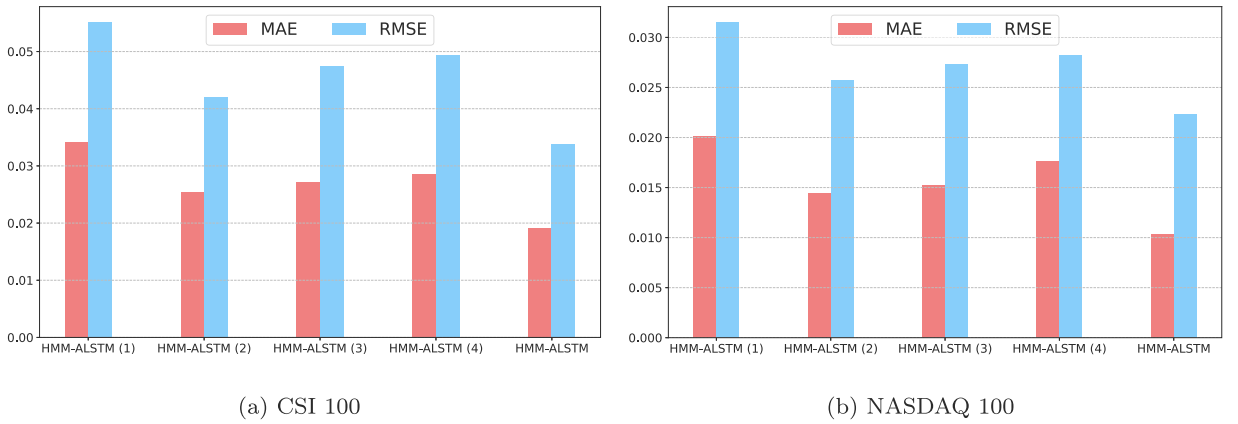


Fig. 4. The performance of HMM-ALSTM on 1-day stock return prediction.

5.3. Ablation studies

To support the effectiveness of the different modules we proposed in HMM-ALSTM, we selected the following variants as a comparison method for implementing ablation study,

- *HMM-ALSTM (1)*: We implement Decoder without the stock state modeling. In other words, this variant only solves the stock prediction problem.
- *HMM-ALSTM (2)*: In this variant, the attention mechanism is only used in Decoder part.
- *HMM-ALSTM (3)*: This variant only utilizes attention mechanism in Encoder part.
- *HMM-ALSTM (4)*: We further remove the attention mechanism in both Encoder and Decoder part in this variant.

We compare the performance of the HMM-ALSTM and its variants. As the data sets from the same stock market show similar results, we only consider CSI 100 and NASDAQ 100 data sets, for conciseness. All the metrics of the 1-day stock movement and return rate prediction are calculated. The results are presented in Figs. 3–4. It can be seen from the results that both of the stock state modeling and attention mechanisms in Encoder and Decoder contribute to the superior performance of the proposed HMM-ALSTM. Specifically, the HMM-ALSTM (1) performs the worst. Compared with HMM-ALSTM, the Accuracy reduces by 2.57% and 2.36% on CSI 100 and NASDAQ 100 respectively, and the MAE increases by 7.81% and 9.51%. This is probably due to the loss of global information in the stock market without the stock state modeling task. Meanwhile, the prediction of return rate is a regression task, which makes it more difficult for the model to effectively capture information and accurately predict ground truth. The results show that the stock state modeling task has a more significant role in promoting the performance of return prediction.

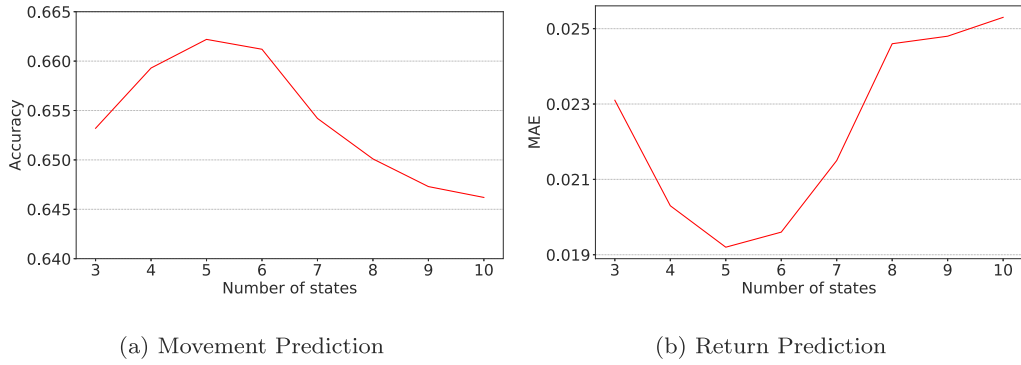


Fig. 5. The performance of different stock states on 1-day stock prediction.

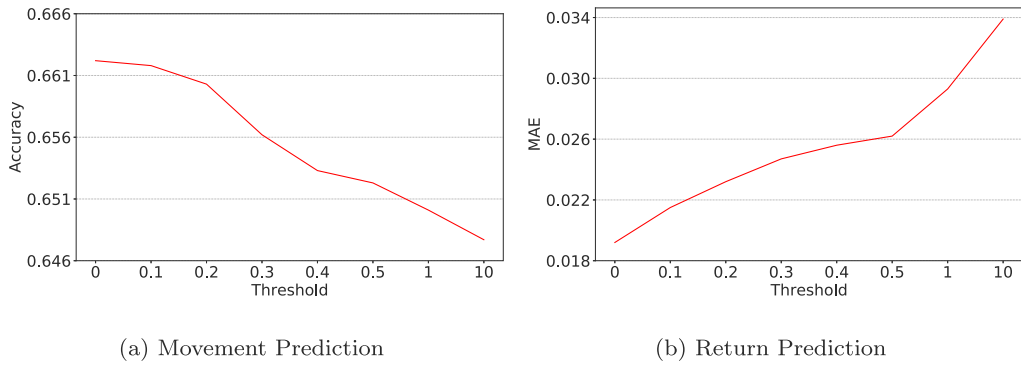


Fig. 6. The performance of different threshold of $Loss_1$ on 1-day stock prediction.

Besides, as the structure of the attention mechanism decreases, the performance of HMM-ALSTM becomes worse as expected. This reflects that the introduction of the attention mechanism helps smooth non-stationary and noisy stock time series. More precisely, the attention mechanism may eliminate noise by assigning weights to data at each time step. Furthermore, HMM-ALSTM (2) performs better than HMM-ALSTM (3) on all metrics. Generally, the deeper layers in the neural network extract more abstract information. In HMM-ALSTM, Decoder extracts high-level information and output the hidden states of HMM. Intuitively, the hidden states effectively integrate the relevant information of other stocks in the entire market. Consequently, the attention mechanism in the Decoder adaptively captures high-level information about the stock and the entire market from the historical state of the stock, thereby contributing to the improvement of the performance of the stock prediction task. This ablation study proves the effectiveness of capturing the stock state patterns and supports the conclusion in Section 5.2.

5.4. Hyperparameter experiment

The ablation study has demonstrated how the stock state modeling task improves the performance stock prediction task. In this section, we conduct hyperparameter experiments to show the influence of the number of stock states and the fitting degree of stock state modeling on the performance of our model. The experiments are also testing the accuracy of stock movement and return rate prediction on CSI 100 and NASDAQ 100 data sets.

We visualize the performance of the proposed HMM-ALSTM with the increase of the number of stock states in Fig. 5, which is also the number of hidden states preset by HMM. There is an interesting relationship between the number of stock states and the performance of the stock prediction task. For our data, the result shows that the ideal number of stock states is 5. When the number of states is less than 5, its improvement to the stock prediction task is outstanding. As the number of stock statuses exceeds 5, the effect of this improvement gradually weakens. At the same time, the extreme number of stock states has a negative impact on model performance. This indicates that the number of stock states beyond a reasonable range not only makes the fitting of stock state modeling difficult, but also introduces redundant noise information to the stock price predictor.

To verify the influence of the fitting degree on the performance of the model, we set the threshold for $Loss_1$ to control the fitting degree of the stock state modeling task, as shown in Fig. 6. It is easy to see that the better the stock state modeling fits, the better

HMM-ALSTM performs. The reason may be that the model can achieve the best effect only when the loss function $Loss_1$ is fully learned when the value of the loss function itself is small. Consequently, a threshold of 0 has the best performance. Specifically, as the threshold increases, the performance of the model decreases faster, which may be because the stock state modeling with low fitness degree misleads learning of the hidden state of LSTM. This experiment also shows the stock state modeling module is indispensable.

6. Discussion

The theoretical contributions of this paper fall within three main aspects. Firstly, we are the first to utilize HMM and neural networks hybrid model to solve stock prediction problem. We propose a novel hybrid structure called HMM-ALSTM to model the stock state and estimate the stock price simultaneously. Besides, we develop a maximum likelihood estimation loss function for the training of stock state modeling and utilize the multi-task learning method and trained weight to combine two tasks. Finally, We also introduce a semi-supervised training procedure, which can dynamically train different modules according to the accuracy of different tasks during the training process. The value of the study lies in the excellent performance of HMM-ALSTM in both stock movement prediction and return rate prediction, which is proved in the abundant experiment results. Meanwhile, the robustness of our method is shown in ablation study and hyperparameter experiment. So the proposed method can maintain long-term stable performance in the fluctuating and noisy stock markets. Moreover, the learned stock states have the potential to contribute in explaining stock movements, and help investors make corresponding decisions.

7. Conclusions

In this paper, we study the classic issue of stock movement forecasting based on the stock time series. To obtain a more stable and effective hidden state in the training stage, we carefully design the semi-supervised state-guiding framework for the LSTM model. This framework possesses a better understanding of the complex trend generating mechanism of stock time series, and we regard the explored pattern as an organic component of the hidden mechanism. In detail, we first propose a novel hybrid structure called HMM-ALSTM to model the stock state and estimate the stock price simultaneously. Further, we develop a maximum likelihood estimation loss function for the training of stock state modeling. For modeling the stock state, the output of the Decoder is the observation state of the hidden Markov model. A standard Multi-Layer Perceptron (MLP) is used to produce the posterior probability of the stock state. Finally, the multi-task learning method is designed to estimate the stock state modeling and stock prediction in parallel. We also introduce a Semi-supervised training procedure, which can dynamically train different modules according to the accuracy of different tasks during the training process. To demonstrate the effectiveness of our proposed model, we choose six popular stock indexes including CSI (CSI 100, 300, and 500), GEI, NASDAQ, and NYSE on the stock market to build our data sets. Sufficient experiment results on the data sets have shown that our model achieved better performance in the tasks of predicting stock movement and return rate. Meanwhile, we make a detailed analysis of the modeled stock state in our hyperparameter experiment, which proves the necessity and effectiveness of our motivation. Our stable and accurate predicting approach is able to help investors make reasonable investments with a less risky future. In the future, we will further explore the model learning of the driving force (state) behind complex markets to propose a more concise and general forecasting scheme.

CRedit authorship contribution statement

Junji Jiang: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Likang Wu:** Methodology, Validation, Writing – original draft. **Hongke Zhao:** Writing – review & editing, Project administration, Funding acquisition. **Hengshu Zhu:** Writing – review & Editing. **Wei Zhang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ipm.2023.103328>. The performance of 3-day and 7-day prediction on HMM-ALSTM and all the baselines is shown in supplementary material.

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